

Overview

This report reviews vulnerabilities affecting the stability of the U.S. financial system related to valuation pressures, borrowing by businesses and households, financial-sector leverage, and funding risks. It also highlights several near-term risks that, if realized, could interact with these vulnerabilities.

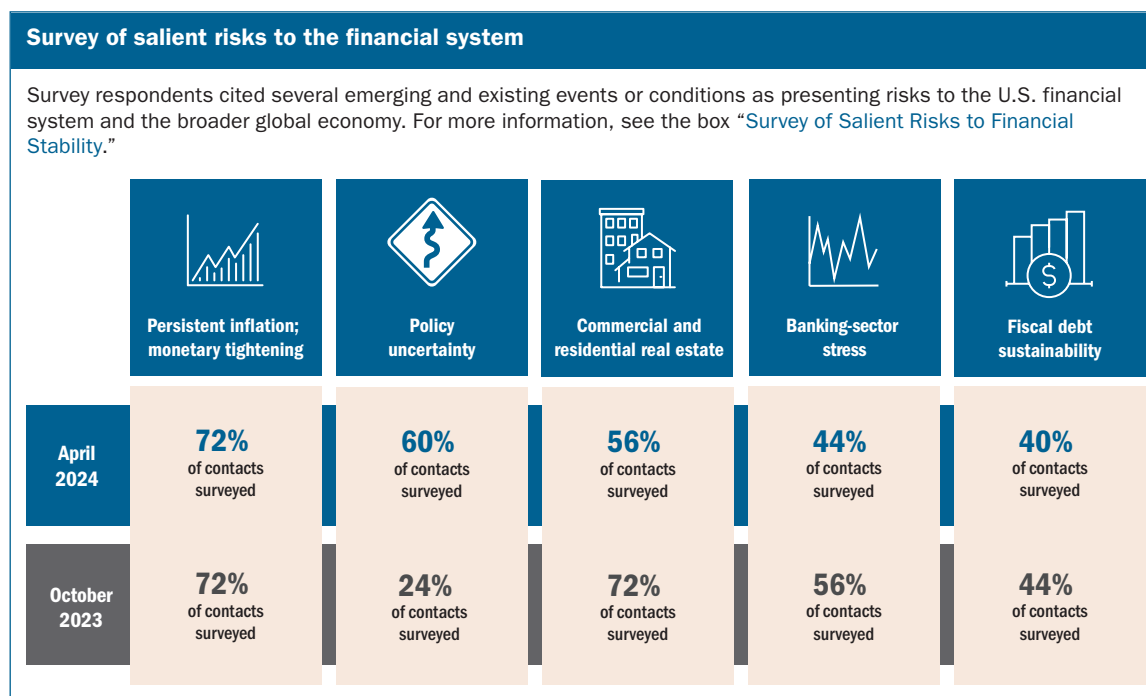
A summary of the developments in the four broad categories of vulnerabilities since the October 2023 *Financial Stability Report* is as follows:

Overview of financial system vulnerabilities			
 Asset valuations	 Borrowing by businesses and households	 Leverage in the financial sector	 Funding risks
• Equity price-to-earnings ratios moved to the upper end of their historical distributions. • Corporate bond spreads fell to levels that are low relative to their historical averages. • Prices of residential real estate remained high relative to fundamentals. • Prices of commercial real estate declined amid deteriorating fundamentals.	• The ratio of total private debt to gross domestic product (GDP) declined further, approaching its historical average. • The business debt-to-GDP ratio remained high, but business debt continued to decline in real terms amid subdued risky debt issuance. Firms' ability to service their debt remained robust. • Household debt was at modest levels relative to GDP and concentrated among prime-rated borrowers.	• The banking system remained sound and resilient, with risk-based capital ratios well above regulatory requirements. • Nonetheless, some banks continued to face sizable fair value losses on some fixed-rate assets held on their balance sheets. • Leverage increased from already elevated levels at the largest hedge funds. • Broker-dealer leverage remained near historically low levels.	• Most domestic banks maintained high levels of liquid assets and stable funding. • However, concerns over uninsured deposits and other factors continued to generate funding pressures for a subset of banks. • Structural vulnerabilities persisted at money market funds, some other mutual funds, and stablecoins. • Life insurers continued to hold a high share of illiquid and risky assets.

1. **Asset valuations.** Valuations rose further to levels that were high relative to fundamentals across major asset classes. Equity prices grew faster than expected earnings, pushing the forward price-to-earnings ratio to the upper end of its historical distribution. Corporate bond spreads narrowed and currently stand at levels that are low relative to their long-run averages. Residential property prices remained high relative to fundamentals and prices continued to rise in recent months. Prices of commercial real estate (CRE) declined amid weak demand for office properties (see Section 1, [Asset Valuations](#)).
2. **Borrowing by businesses and households.** The balance sheets of nonfinancial businesses and households remained solid, as the ratio of total private debt to gross domestic product (GDP) declined further, approaching its historical average. Although business debt remained high when measured relative to GDP (or to business assets for publicly traded corporations), business debt declined in real terms throughout last year. Firms' ability to service their debt remained robust owing to strong earnings and low borrowing costs on existing debt. Household debt remained at modest levels relative to GDP, and most of that debt is owed by households with strong credit histories or considerable home equity (see Section 2, [Borrowing by Businesses and Households](#)).
3. **Leverage in the financial sector.** The banking sector remained sound and resilient overall, and most banks continued to report capital levels well above regulatory requirements. Nevertheless, fair value losses on fixed-rate assets remained sizable for some banks, and some banks with concentrated exposure to loans backed by commercial real estate properties experienced stress. Outside the banking sector, available data suggest that hedge fund leverage grew to historic highs, driven primarily by borrowing by the largest hedge funds. Leverage at life insurance companies remained around its median, while they continued to take on credit and liquidity risk. Broker-dealer leverage remained near historical lows (see Section 3, [Leverage in the Financial Sector](#)).
4. **Funding risks.** Liquidity at most domestic banks remained ample, with limited reliance on short-term wholesale funding. Nevertheless, some banks continued to face funding strains, likely owing to vulnerabilities associated with high levels of uninsured deposits, declines in the fair value of assets, and elevated exposure to CRE. Structural vulnerabilities remained in other short-term funding markets. Prime and tax-exempt money market funds (MMFs), as well as other cash-investment vehicles and stablecoins, remained vulnerable to runs. Bond and loan funds that hold assets that can become illiquid during periods of stress remained susceptible to large redemptions. In addition, life insurers continued to rely on a higher-than-average share of nontraditional liabilities (see Section 4, [Funding Risks](#)).

This report also discusses potential near-term risks, based in part on the most frequently cited risks to U.S. financial stability as gathered from outreach to a wide range of researchers, academics, and market contacts conducted from late January through the end of March (discussed in the box “[Survey of Salient Risks to Financial Stability](#)”). The risk of persistent inflationary pressures leading to a more restrictive than expected monetary policy stance remained the most frequently cited risk, mentioned by nearly three-fourths of survey participants. The share of survey participants mentioning policy uncertainty as a risk to the financial system stood at just under two-thirds, significantly higher than in the October report. Over half of all survey participants mentioned the potential effect of large realized losses on CRE and residential real estate, down from three-fourths of all participants in the previous survey. Rounding out the top five, risks associated with the reemergence of banking-sector stress and with fiscal debt sustainability in advanced economies continued to feature prominently.

In addition, the report also contains the box “[The Bank Term Funding Program](#),” which describes the role the program played in providing funding to the banking system beginning with its inception in response to the March 2023 banking-sector stresses up until it ceased extending new loans on March 11, 2024.



1 | Asset Valuations

Asset valuations increased to elevated levels relative to fundamentals

Since the October report, equity valuations increased further. Valuations in corporate bond markets also appeared stretched as corporate credit spreads, the difference in yields on corporate bond and yields on similar-maturity Treasury securities, narrowed since the previous report, falling to levels in the lower range of their historical distributions. Liquidity in short-term Treasury markets remained low by historical standards, although market liquidity was consistent with elevated measures of interest rate volatility. Nonetheless, Treasury market liquidity conditions could amplify the impact of shocks on asset valuations.

Residential real estate valuations remained near the peak levels seen in the mid-2000s. CRE market conditions continued to deteriorate, especially for the office sector, and prices continued to decline against a backdrop of high vacancy rates and weakening rents. Farmland prices were historically elevated relative to rents, reflecting limited inventories of land.

Table 1.1 shows the sizes of the asset markets discussed in this section. The largest asset markets are those for equities, residential real estate, Treasury securities, and CRE.

Treasury yields decreased slightly and remain high relative to the past 15 years

Yields on Treasury securities decreased slightly since the October report but remained close to their highest levels over the past decade and a half (figure 1.1). A model-based estimate of the nominal Treasury term premium—a measure of the compensation that investors require to hold longer-term Treasury securities rather than shorter-term ones—remained low relative to its long-run history despite edging up through March (figure 1.2). While interest rate volatility implied by options declined a touch, it remained elevated by historical norms (figure 1.3). This volatility reflected, in part, uncertainty about the economic outlook and the associated path of monetary policy, which likely heightened the sensitivity of Treasury yields to news about output growth, inflation, and the supply of Treasury securities.

Table 1.1. Size of selected asset markets

Item	Outstanding (billions of dollars)	Growth, 2022:Q4–2023:Q4 (percent)	Average annual growth, 1997–2023:Q4 (percent)
Equities	57,175	22.2	9.2
Residential real estate	56,415	3.6	6.2
Treasury securities	26,227	10.0	8.2
Commercial real estate	22,518	–6.3	6.4
Investment-grade corporate bonds	7,533	5.4	8.1
Farmland	3,420	7.7	5.8
High-yield and unrated corporate bonds	1,631	–2.6	6.2
Leveraged loans ¹	1,397	–1.1	13.2
Price growth (real)			
Commercial real estate ²		–1.3	3.1
Residential real estate ³		2.1	2.7

Note: The data extend through 2023:Q4. Growth rates are measured from Q4 of the year immediately preceding the period through Q4 of the final year of the period. Equities, real estate, and farmland are at nominal market value; bonds and loans are at nominal book value.

¹ The amount outstanding shows institutional leveraged loans and generally excludes loan commitments held by banks. For example, lines of credit are generally excluded from this measure. Average annual growth of leveraged loans is from 2000 to 2023:Q4, as this market was fairly small before then.

² One-year growth of commercial real estate prices is from December 2022 to December 2023, and average annual growth is from December 1999 to December 2023. Both growth rates are calculated from equal-weighted nominal prices deflated using the consumer price index (CPI).

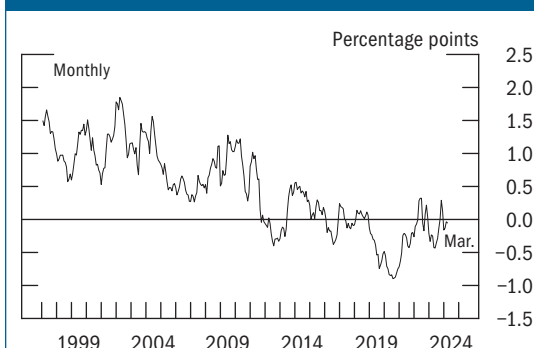
³ One-year growth of residential real estate prices is from December 2022 to December 2023, and average annual growth is from December 1998 to December 2023. Nominal prices are deflated using the CPI.

Source: For leveraged loans, PitchBook Data, Leveraged Commentary & Data; for corporate bonds, Mergent, Inc., Fixed Income Securities Database; for farmland, Department of Agriculture; for residential real estate price growth, CoreLogic, Inc.; for commercial real estate price growth, CoStar Group, Inc., CoStar Commercial Repeat Sale Indices; for all other items, Federal Reserve Board, Statistical Release Z.1, "Financial Accounts of the United States."

Figure 1.1. Nominal Treasury yields remained close to the highest levels in the past 15 years

Source: Federal Reserve Board, Statistical Release H.15, "Selected Interest Rates."

Figure 1.2. An estimate of the nominal Treasury term premium remained relatively low



Source: Department of the Treasury; Wolters Kluwer, Blue Chip Financial Forecasts; Federal Reserve Bank of New York; Federal Reserve Board staff estimates.

Figure 1.3. Interest rate volatility fell slightly but continued to be elevated by historical norms



Source: For data through July 13, 2022, Barclays and S&P Global; for data from July 14, 2022, onward, ICAP, Swaptions and Interest Rate Caps and Floors Data.

Measures of equity market valuations rose further from already high levels

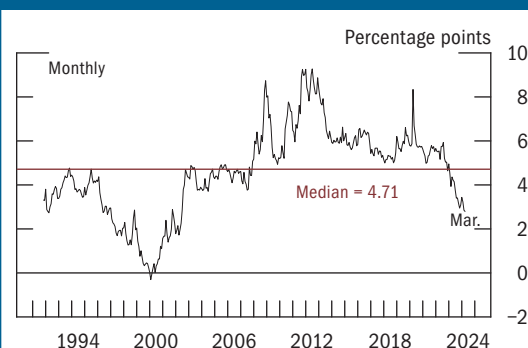
The ratio of prices to expected 12-month earnings, or the price-to-earnings ratio, increased since the October report and currently sits in the upper end of its historical distribution since 1989 (figure 1.4). The difference between the forward earnings-to-price ratio and the real 10-year Treasury yield—a measure of the additional return that investors require for holding stocks relative to risk-free bonds (the equity premium)—declined, on net, since the October report and currently stands well below its historical median (figure 1.5).² Equity market volatility was subdued,

Figure 1.4. The price-to-earnings ratio of S&P 500 firms increased to levels further above its historical median



Source: Refinitiv, Institutional Brokers' Estimate System, North American Summary & Detail Estimates, Level 2, Current & History Data, Adjusted and Unadjusted, <https://www.lseg.com/en/data-analytics/financial-data/company-data/ibes-estimates>.

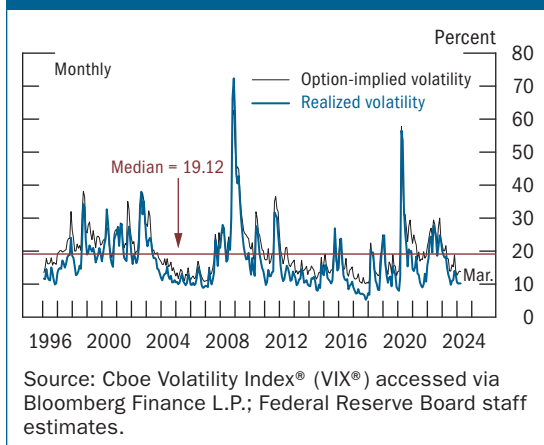
Figure 1.5. An estimate of the equity premium fell further below its historical median



Source: Refinitiv, Institutional Brokers' Estimate System, North American Summary & Detail Estimates, Level 2, Current & History Data, Adjusted and Unadjusted, <https://www.lseg.com/en/data-analytics/financial-data/company-data/ibes-estimates>.

² This estimate is constructed based on expected corporate earnings for 12 months ahead. Alternative measures of the equity premium that incorporate longer-term earnings forecasts suggest more elevated equity valuation pressures.

Figure 1.6. Volatility in equity markets decreased to levels slightly below the historical median



and option-implied volatility remained in the lower quarter of its historical distribution (figure 1.6).

Spreads in corporate debt markets narrowed to low levels

Yields for both investment- and speculative-grade bonds fell a bit since the October report and currently stand near the median of their respective historical distributions (figure 1.7). While the decline in corporate bond yields was modest, it nevertheless outpaced that of comparable-maturity Treasury securities, and, as a result, corporate bond spreads narrowed to levels that are low relative to their historical distributions (figure 1.8). The excess bond premium—a risk premium measure that captures the gap between corporate bond spreads and expected credit losses—remained near its historical mean (figure 1.9). Market-based forecasts of credit quality (one-year-ahead default probabilities) of nonfinancial firms have mildly improved since the October report but remain somewhat elevated by historical standards for speculative-grade issuers.

Figure 1.7. Corporate bond yields fell slightly to levels near their historical medians

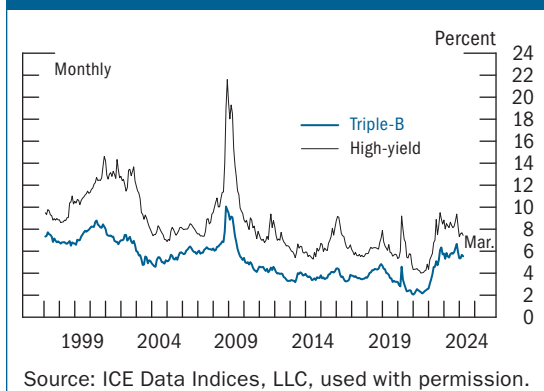


Figure 1.8. Corporate bond spreads narrowed to low levels relative to their historical distributions

